

Experiment 2

Designing and Implementing ETL Processes using Microsoft Power BI

AIM:

To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

OBJECTIVES:

- To extract data from real-world business datasets using Microsoft Power BI.
- To transform and clean the extracted data using Power Query Editor.
- To apply data transformation techniques such as filtering, merging, removing duplicates, handling missing values, and changing data types
- To load the transformed data into a unified analytical model.

TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Sample datasets: Global Superstore 2018.xlsx , Employee_DB.xlsx and a sample SQL database

DATASET USED: Global Superstore 2018.xlsx, Employee_DB.xlsx, Transactions.sql

THEORY:

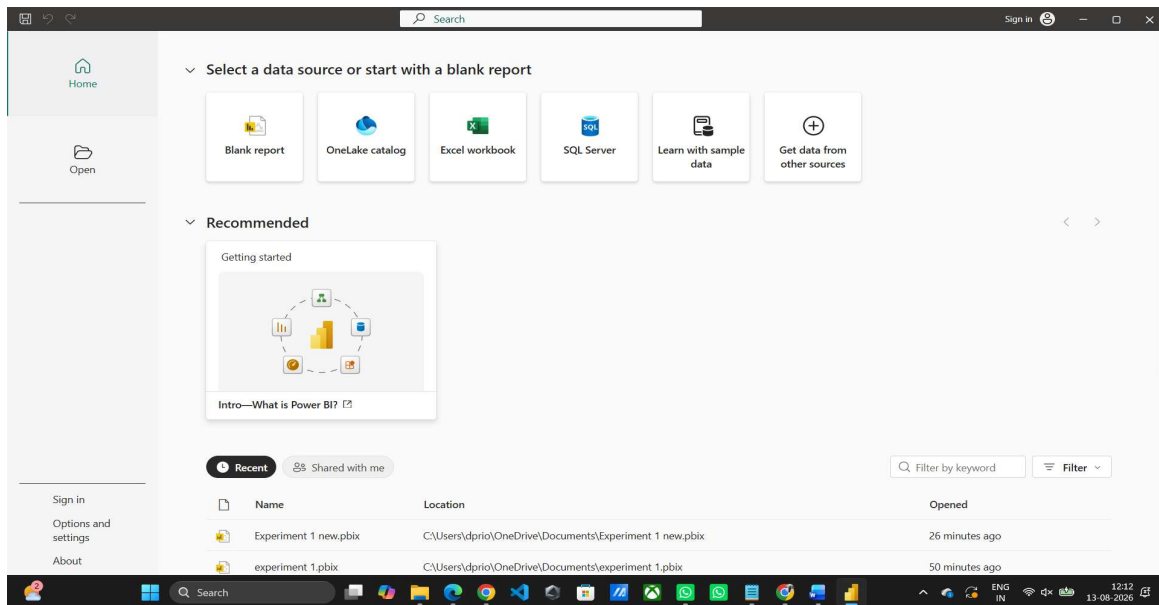
1. **ETL** (Extract, Transform, Load) is one of the most important processes in Business Intelligence. It enables organizations to collect data from source systems, transform it into a clean and consistent format, and load it into a centralized environment for analysis.
2. **Extract Phase:** The extraction phase involves collecting data from source systems. In this experiment, data is extracted from the Orders, Returns, and People worksheets available in the Global Superstore dataset.
3. **Transform Phase:** The transformation phase includes data cleaning, handling missing values, correcting data types, creating derived columns, filtering invalid records, and integrating multiple datasets.
4. **Load Phase:** The transformed data is loaded into the Power BI data model where relationships can be established and analytical measures can be created using DAX.
5. **Power Query Editor** serves as the ETL engine in Power BI and records each transformation step as an Applied Step, making the workflow transparent and reproducible.

PROCEDURE:

Part A: Extract-Connect to Dataset

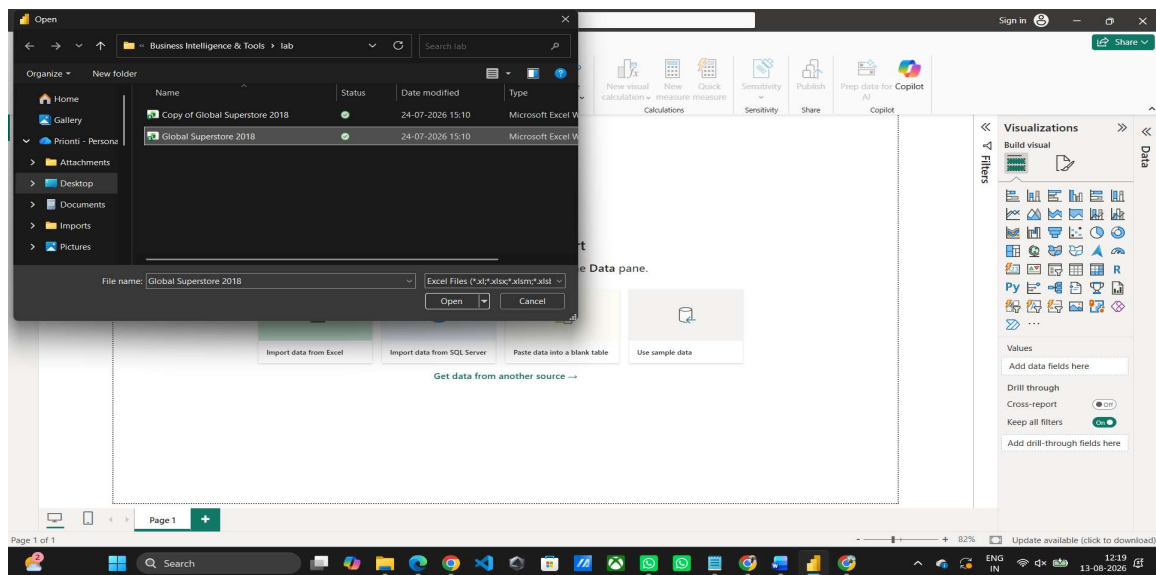
Step 1: Download the Global Superstore Dataset from Kaggle.

Step 2: Open Power BI Desktop.



Home > Get Data > Excel Workbook

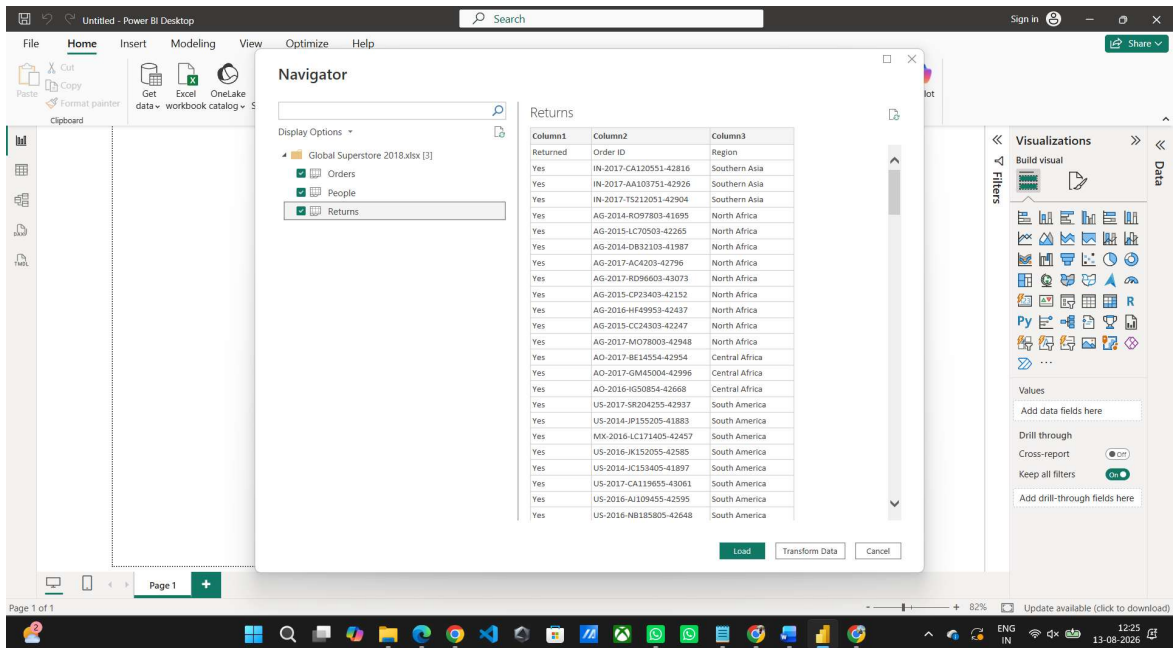
Select Global Superstore Dataset.xlsx



Click Transform Data.

Step 3: Import the following worksheets:

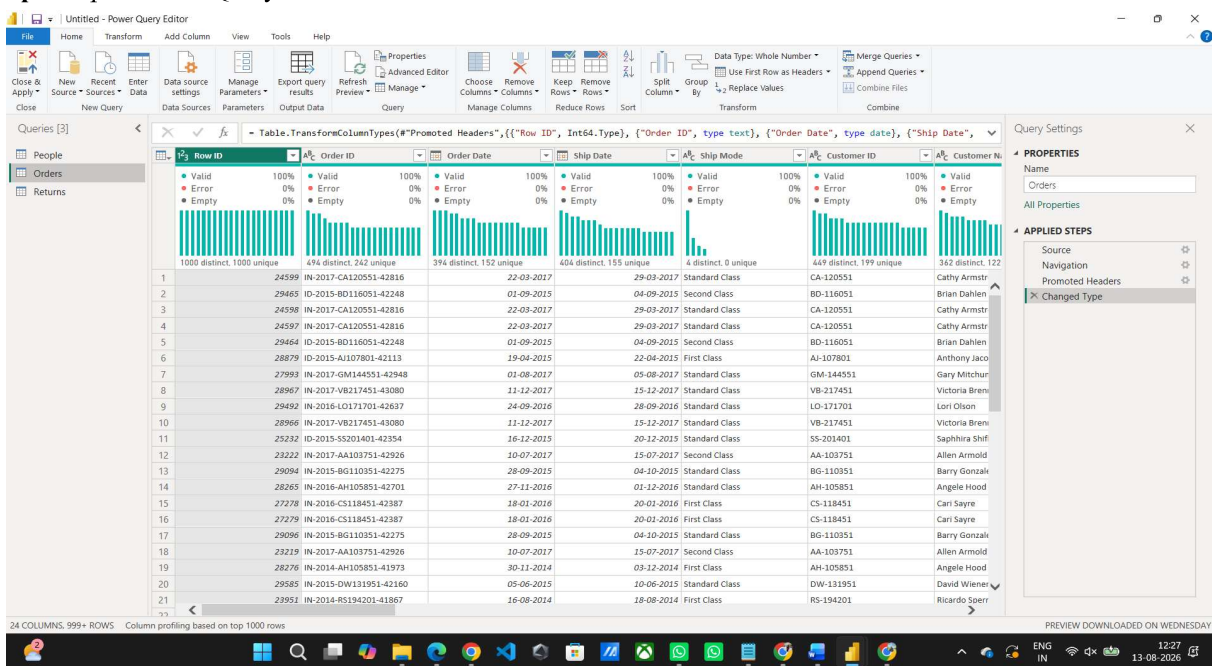
- Orders
- Returns
- People



Verify that all imported tables appear in the Queries pane.

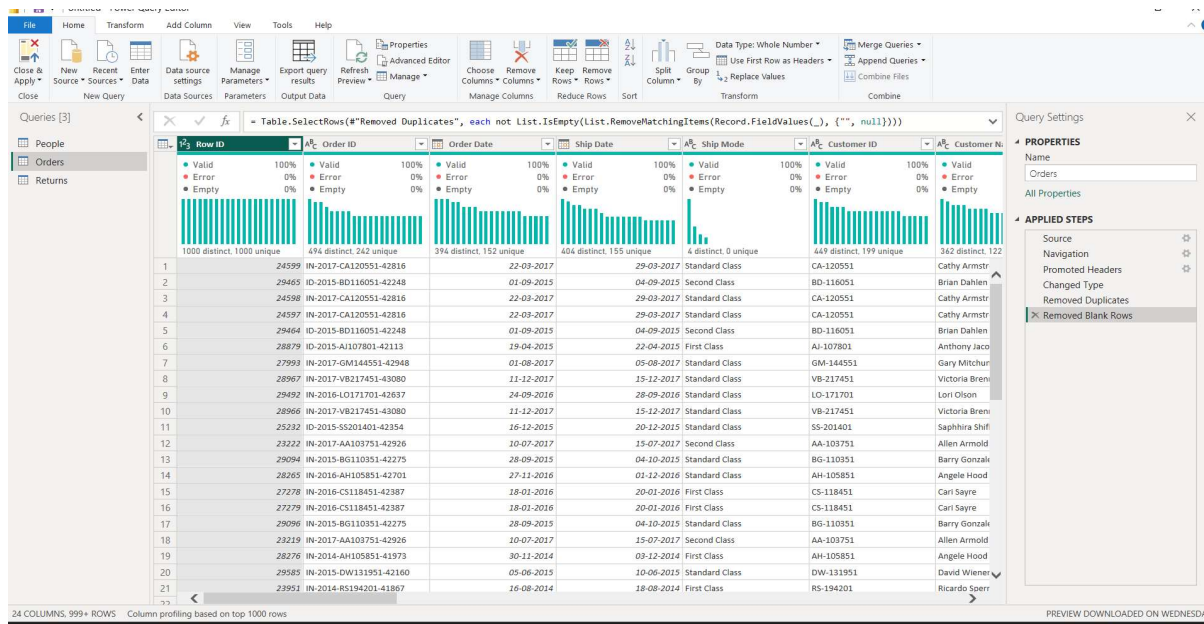
Part B: Transform-Power Query Editor

Step 4: Open Power Query Editor.

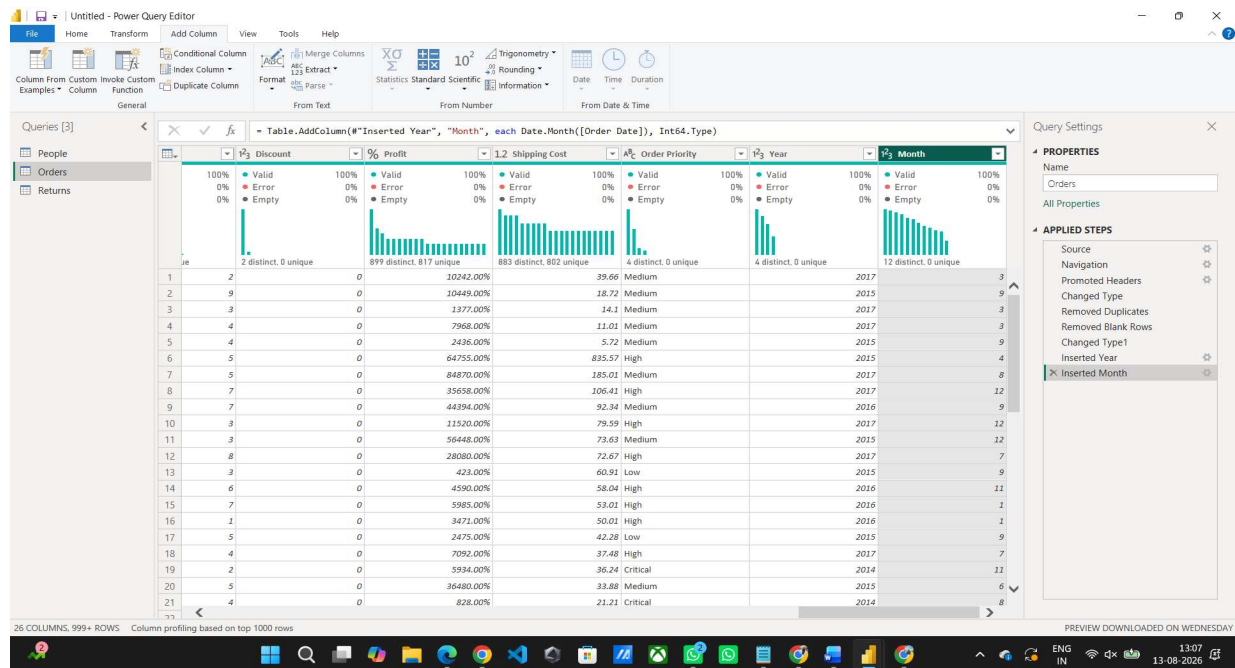


Step 5: Apply the following transformations to the Orders table:

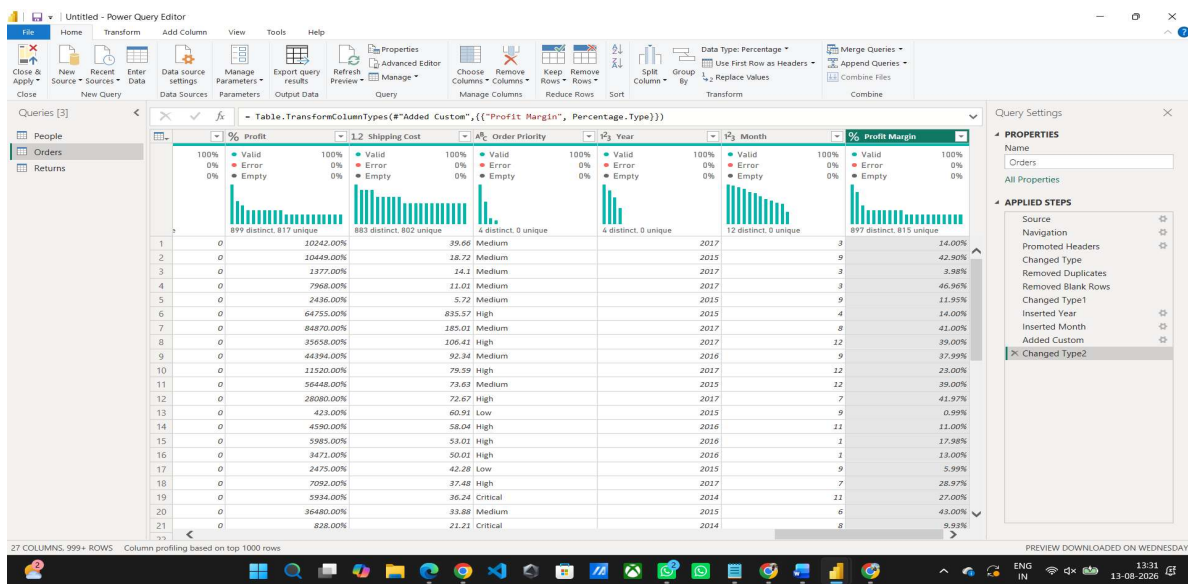
- Remove duplicate records
- Remove blank rows



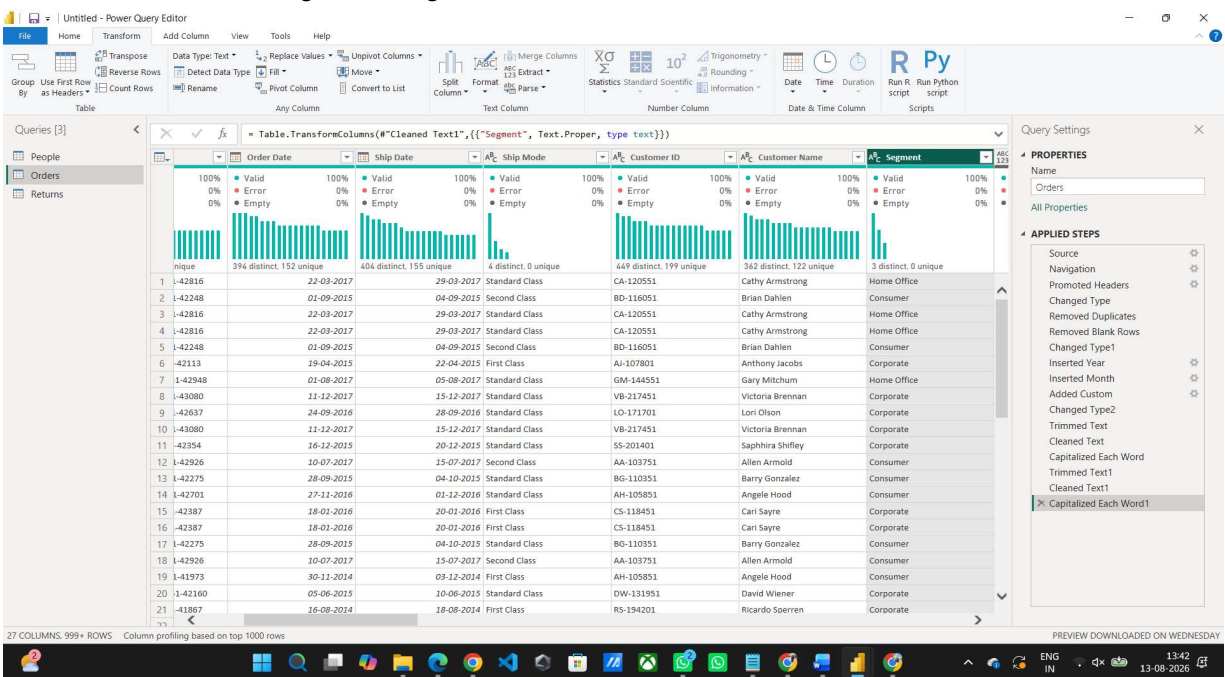
- Verify and correct data types
- Rename columns where required
- Extract Year and Month from Order Date



- Create a custom Profit Margin column

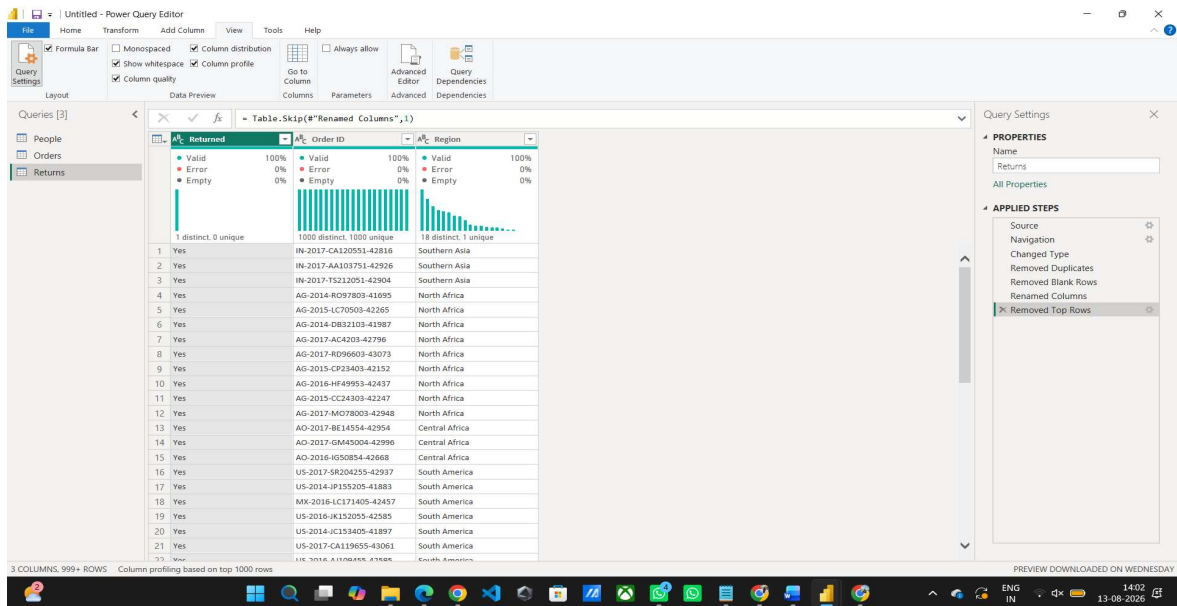


• Standardize Region and Segment values



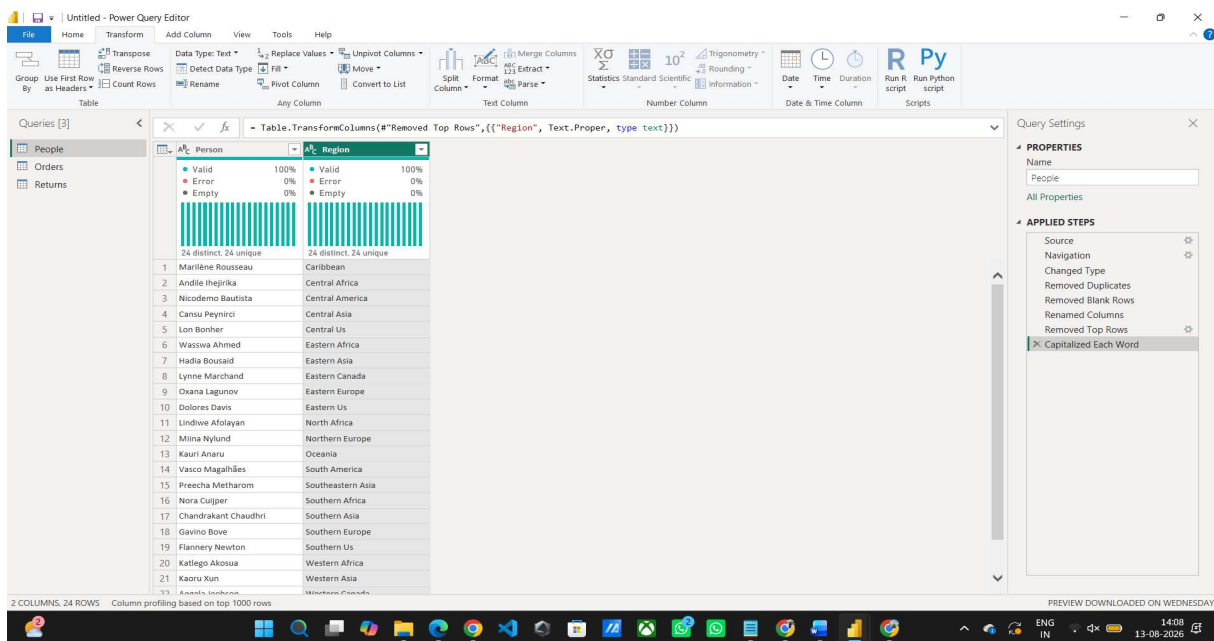
Step 6: Transform the Returns table:

- Remove duplicate Order IDs
- Check for missing values
- Standardize Returned field values

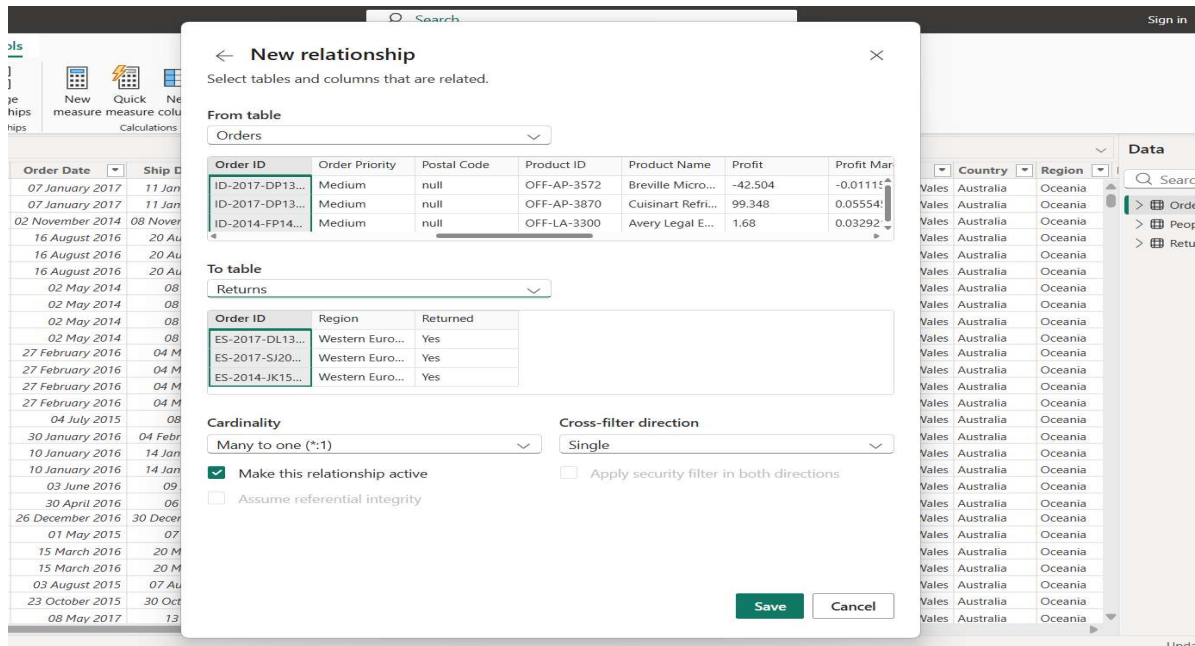


Step 7: Transform the People table:

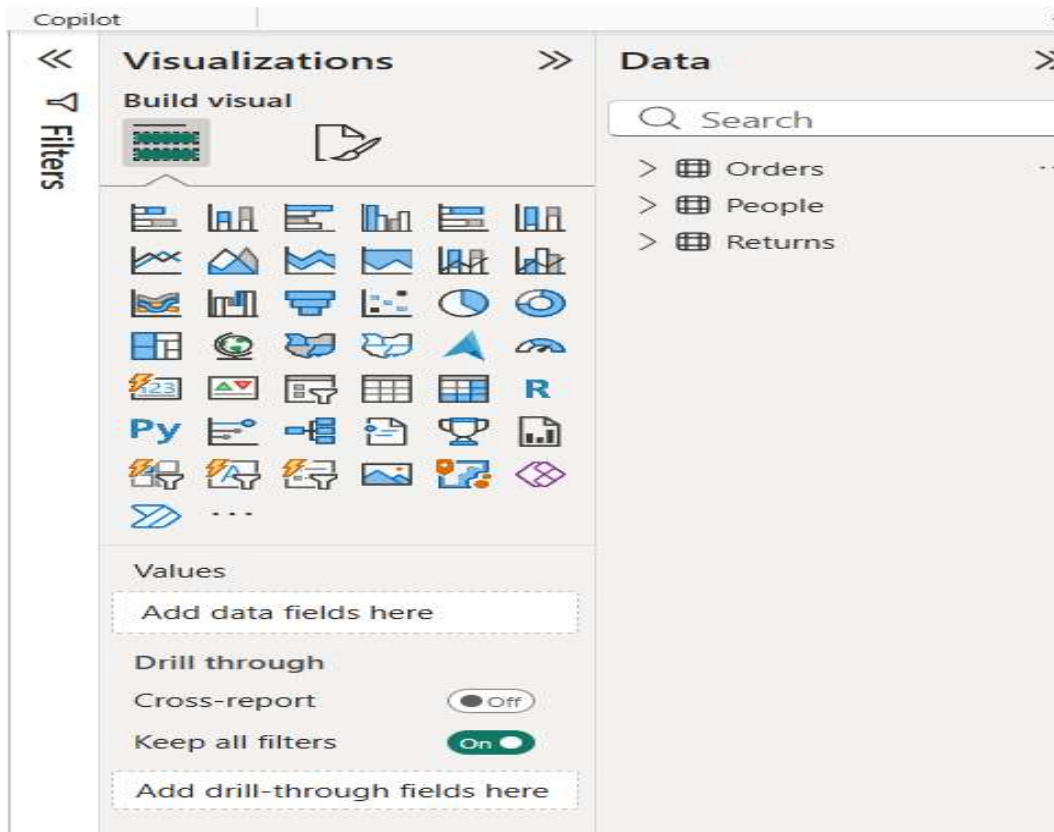
- Validate regional assignments
- Remove unnecessary columns
- Verify consistency of region names



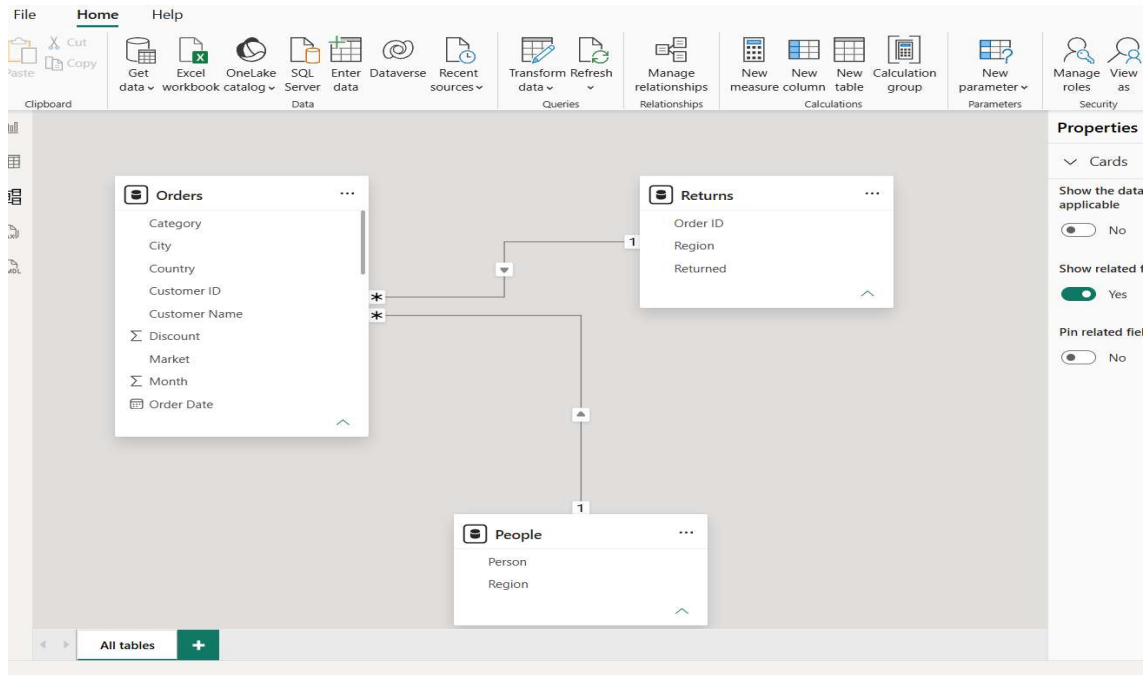
Step 8: Merge Orders and Returns tables using Order ID.



Step 9: Review all Applied Steps and click Close & Apply.



Step 10: Navigate to Model View.



Orders $\leftrightarrow$ Returns (Order ID)

Orders $\leftrightarrow$ People (Region)

Manage relationships

+ New relationship

⚡ Autodetect

Edit

Delete

Filter

✓ From: table (column) ↑	Relationship	To: table (column)	Status
✓ Orders (Order ID)	* — ◀ — 1	Returns (Order ID)	Active ...
✓ Orders (Region)	* — ◀ — 1	People (Region)	Active ...

Total Sales = SUM(Orders[Sales])

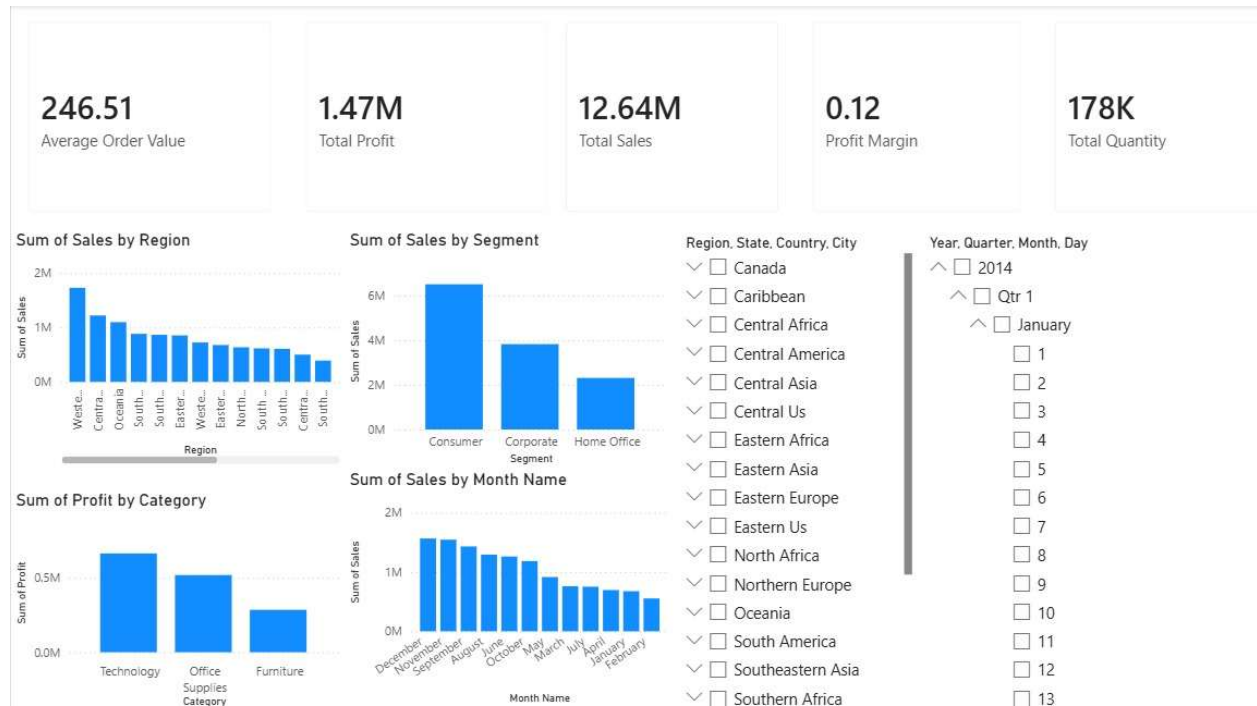
Total Profit = SUM(Orders[Profit])

$$\text{Profit Margin} = \text{DIVIDE}(\text{SUM}(\text{Orders}[\text{Profit}]), \text{SUM}(\text{Orders}[\text{Sales}]))$$

Total Orders = DISTINCTCOUNT(Orders[Order ID])

Step 13: Create a simple report visual using fields from related tables.

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RESULT:

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

LEARNING OUTCOMES:

- Explain the ETL process and its role in Business Intelligence systems.
- Extract data from real-world business datasets using Power BI.
- Apply transformation techniques including cleaning, filtering, and merging.
- Create relationships and analytical measures in Power BI
- Validate transformed datasets for reporting and decision-making.